



FedQR: Communication-Efficient Federated Learning via QR Factorization

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Abstract. While federated learning is a state-of-the-art framework for distributed privacy-preserving machine learning, it faces significant challenges in scenarios with non-homogeneous data. Data heterogeneity causes training instability and slows convergence. Existing reparameterization and knowledge distillation solutions introduce new challenges, such as controlling client drift and avoiding knowledge interference. In this paper, we propose a federated learning method based on QR factorization (FedQR) to address these issues. FedQR reparameterizes local model updates by projecting gradients onto the orthogonal basis of the global weights. This approach effectively mitigates client drift while seamlessly retaining the knowledge already acquired by the global model. Our extensive experiments on common federated learning vision and text datasets demonstrate that FedQR can improve communication cost by $7.6\times$ and increase accuracy by up to 10% compared to state-of-the-art methods, particularly in non-i.i.d. scenarios. Extended evaluations on real-world feature shift datasets further confirm the robustness and performance of our method, with accelerated convergence of up to $20\times$.

Keywords: Federated learning communication · Heterogeneous data · QR factorization · Reparameterization

1 Introduction

The recent advancements in the Internet of Things (IoT) have led to an explosive growth of data, which has significantly contributed to the success of deep learning in delivering intelligent healthcare services [17], precision agriculture [6], and smart transportation systems [19]. Traditional deep learning methods

typically train models on centrally collected data from devices, which can compromise users' data privacy. Alternatively, federated learning has emerged as a new paradigm that allows distributed learning across multiple devices without the need to share raw data. It enables privacy preserving learning while benefiting from the computational power, communication capabilities, and diverse datasets of devices.

The classical Federated Average (FedAvg) method performs well when data are independent and identically distributed (i.i.d) [20]. However, in practice, each device independently gathers its data, resulting in a lack of independence and identical distribution (non-i.i.d) data across devices. This variability significantly degrades training convergence, thus slowing down communication. The study in [16] has identified the presence of heterogeneous data to cause local models' direction to move far from the global model. To address this limitation, one approach is to utilize model reparametrization technique to control the *deviation* of local models from the global model. While existing methods, such as FedProx [16], FedDyn [1], and SCAFFOLD [11] has improved the convergence of the vanilla FedAvg, they often suffer from degraded performance and introduce hyperparameters that are highly sensitive, especially in low participation settings. More importantly, finding the balance in local model direction appear to be very challenging in highly heterogeneous setting.

On the other hand, Federated knowledge distillation (FedKD) methods take a different approach by transferring knowledge from a teacher model (usually the global model) to a student model (local model). For instance, FedMD combines transfer learning and knowledge distillation, allowing each client to possess its data distribution and model architecture [15]. Authors in [2] propose to transfer knowledge through additional training on shared public data. However, the success of such methods depends on the alignment of data distribution between the proxy and the clients, which, in practice, is often not the case. To mitigate this issue, Zhu et al. proposed an approach based on generative synthetic data [31]. They introduces a lightweight data-free method by incorporating a generative model to learn from the raw feature space, avoiding the local bias effect. Despite their success, FedKD methods have two key issues: (i) they may involve sharing generative or public data, potentially leading to privacy breaches, and (ii) the proxy data might not align with private data, which induces global knowledge forgetting.

Motivated by these observations, we propose a *projected QR factorization for federated learning* (FedQR) to efficiently constrain *client drift* in heterogeneous data, while retaining the essential features learned from the global model. Our approach revolves around introducing a more stable and efficient reparametrization technique with a significant reduction of model weights' dimensions. To accomplish this, we adopt a two-step process. First, we determine a new basis by performing QR factorization on the parameter space of the global model. Next, we project the local gradient onto the orthogonal direction spanned by this basis. Our evaluation on both simulated label skew data using MNIST [13], Fashion-MNIST (FMNIST) [24], and CIFAR-10 [12] datasets, as well as real-world feature

shift data such as Digit-5 and Amazon-review [28] datasets, demonstrates that the proposed method can significantly accelerate communication by up to $20\times$ without the need for learning rate tuning. The introduction of our QR factorization strategy into the objective function also yields to a more stable convergence and higher accuracy.

2 Related Works

2.1 Problem of Non-i.i.d Data in Federated Learning

A key challenge in federated learning for real-world applications is the distributed nature of devices, each having its own data distribution. FedAvg has shown degraded performance in heterogeneous scenarios, mainly because it performs local Stochastic Gradient Descent (SGD), which is effective only under the i.i.d assumption. Consequently, local updates based on SGD optimization can deviate from the global model when data is heterogeneous. Various improvements to FedAvg were proposed to mitigate such drift, including warmup training [29], incorporating a proximal term [16], and using adaptive optimization [22]. While these methods enhance FedAvg’s convergence, they introduce additional hyperparameters and require extensive tuning.

2.2 Model Reparameterization in Federated Learning

Reparameterization methods, such as FedProx [16], rely on full matrix weights, leading to increased bit transfer between clients and the server, as well as heightened computational costs. In response to these challenges, many studies [9, 26, 30] have explored matrix factorization to reduce communication costs and improve generalization performance. In the context of federated continual learning, researchers have adopted local parameter decomposition to mitigate knowledge forgetting and enhance model robustness [4, 27]. This approach prevents the interference of current model updates with previously acquired knowledge, effectively preserving model footprint over time. Additionally, efforts have been made to transfer knowledge between clients using a combination of general parameters and task-specific sparse representations generated through model weight factorization [10]. To the best of our knowledge, our method is the first to introduce an orthogonal projection approach to preserve global model knowledge in non-i.i.d federated learning settings.

2.3 QR Factorization

Let $\mathbf{X}$ be a weight matrix of m rows and n columns. Following the study in [14] which shows that over-parameterize neural network weight has a low intrinsic dimension, the QR factorization of $\mathbf{X}$ decomposes it into the product of an orthogonal matrix $\mathbf{Q}$ and an upper triangular matrix $\mathbf{R}$, such that $\mathbf{X} = \mathbf{QR}$, where $\mathbf{Q} \in \mathbb{R}^{m \times k}$ and $\mathbf{R} \in \mathbb{R}^{k \times n}$ with $k = \min(m, n)$ [23]. Since $\mathbf{Q}$ is an orthogonal matrix, it satisfies the property $\mathbf{Q}^T \cdot \mathbf{Q} = \mathbf{I}$, where $\mathbf{I}$ is the identity

matrix. To compute this decomposition, various methods have been proposed. Here, we utilize the Householder method [7] to decompose our global weight matrix. The Householder method provides an alternative to earlier techniques such as the Gram-Schmidt process [3] for computing the QR factorization of a matrix. While the Gram-Schmidt process has been historically used, its stability and efficiency have been found lacking in certain scenarios. In contrast, the Householder method offers improved stability and efficiency. In essence, the Householder method involves employing orthogonal transformations known as Householder reflectors to reflect a given vector x onto a plane. These reflectors are defined by matrices of the form as follow:

$$H = \mathbf{I} - 2ww^T \quad (1)$$

where H is the Householder reflector, w is a suitable vector and $\mathbf{I}$ is an identity matrix. The resulting upper triangular matrix $\mathbf{R}$ in the QR factorization is obtained through a series of Householder transformations applied successively from H_{m-1} down to H_0 , such that:

$$\mathbf{R} = H_{m-1} \dots H_0 \mathbf{X} = \mathbf{Q}^T \mathbf{X} \quad (2)$$

It's worth noting that while the $\mathbf{Q}$ matrix is not explicitly defined in terms of Householder reflectors, it can be reconstructed using the product of the Householder matrices, as follow:

$$\mathbf{Q} = H_1 H_2 \dots H_{m-2} H_{m-1} \quad (3)$$

This allows for the full reconstruction of the orthogonal matrix $\mathbf{Q}$ from the Householder reflectors.

3 Methodology

3.1 Problem Setup

In the context of federated learning with N participating nodes and their respective loss function $f_i(\mathbf{x})$. The optimization goal is to find $\mathbf{x}$ that can minimize the global function $f(\mathbf{x})$ as follows:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \left[f(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}) \right], \quad (4)$$

The loss function of the worker node at the i -th index with private training data ξ^i sampled from the data distribution $\mathcal{D}_i$ is:

$$f_i(\mathbf{x}) := \mathbb{E}_{\xi \sim \mathcal{D}_i} [F_i(\mathbf{x}, \xi)], \quad (5)$$

In the case of finite data point $\{\mathcal{D}_i^m\}_{m=1}^M$, the loss function can be rewritten as an empirical risk minimization problem:

$$f_i(\mathbf{x}) = \frac{1}{m_i} \sum_{j=1}^{m_i} F_i(\mathbf{x}, \xi_j^i) \quad (6)$$

Each worker node trains locally the model while only sharing weights with the server. Then, during each iteration $t \in T$, all the local weights $\mathbf{x}_{i,t}$ are aggregated to obtain a global weight $\mathbf{x}_t$. In our setting, the data distribution $\mathcal{D}_i$ across nodes can be heterogeneous and the optimal local model $\mathbf{x}_i^*$ can also deviate from the global minimum $\mathbf{x}^*$. In real-world federated learning scenarios involving multiple worker nodes, some can possibly be unavailable or unreliable. The full participation setting in Eq. 4 requires all nodes to be part of the training, which is unrealistic. An alternative approach is partial device participation, where at each round a number of device $\mathcal{S}_t$ is randomly selected from a uniform client distribution $|\mathcal{S}_t|$.

3.2 Details of the Proposed Method

In this section, we present FedQR, a new method that utilizes QR factorization to regulate local gradient updates in federated learning. FedQR aims to enhance stability, effectively curb client drift, and improve model convergence.

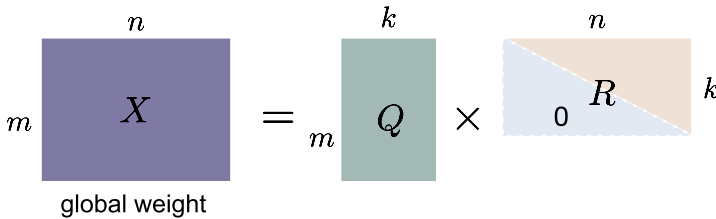


Fig. 1. Global weight decomposition via QR factorization

Our approach focuses on reparameterizing the model weights to effectively control the divergence of the local model from the global model. Unlike FedProx and FedDyn, which operate within the entire global model space, FedQR updates its basis by decomposing the global model parameter using QR factorization. This decomposition enables stability and fast convergence during training.

Method Overview: Considering the data distributions $\mathcal{D}_i$, where $\mathcal{D}_i \neq \mathcal{D}_j \forall i \neq j$, indicating data heterogeneity, at each global round in T , clients train the model based on the previous global model iteration. These local updates can significantly alter the global weights and interfere with the knowledge already acquired. In this study, we attempt to limit the changes in the global model during local updates by projecting onto a subspace orthogonal to the global model. Concretely, given the current gradient of the local loss function $\nabla F_{i,t}(\mathbf{x}, \xi)$ and the previous global weight $\mathbf{x}_{t-1}$, we want to project the current gradient into a space orthogonal to $\text{span}\{\mathbf{X}_{t-1}\}$. Using the Householder reflector in Eq. 3, we

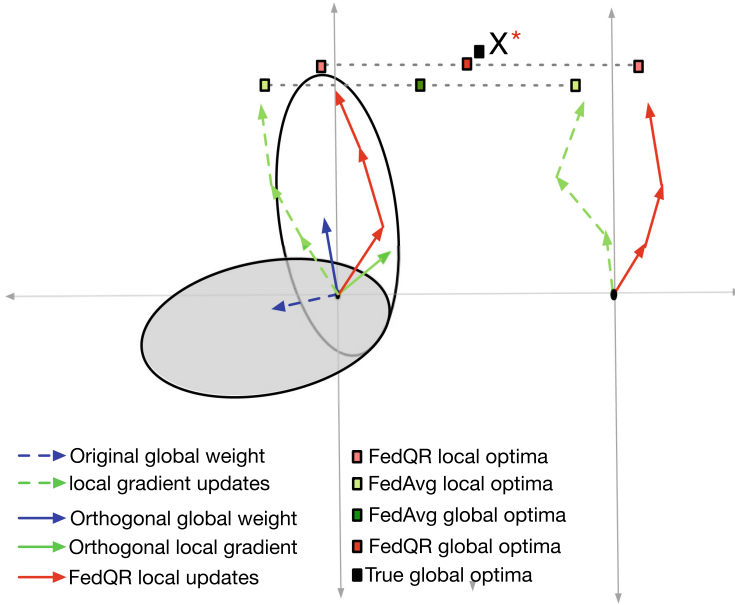


Fig. 2. Visualization of FedQR: Our proposed method begins by deriving a new basis for the original global weights through QR factorization. Next, the local gradient is updated by projecting onto the subspace orthogonal to this new basis. This adjustment enables FedQR to bring the global model closer to the true global optimum, in contrast to FedAvg, which tends to deviate from the true global optimum.

obtain a new orthogonal basis $\mathbf{Q} = H_1 H_2 \dots H_{m-2} H_{m-1}$. Our projection can be formulated as follows:

$$\text{proj}(\mathcal{G}) = \mathbf{Q}^T \nabla F_i(\mathbf{x}, \xi) \quad (7)$$

Finally, the direction of $\nabla F_i(\mathbf{x}, \xi)$ is corrected to our new projected gradient, denoted as $\tilde{g}$, as follows:

$$\tilde{g} = \nabla F_i(\mathbf{x}, \xi) - \text{proj}(\mathcal{G}) \quad (8)$$

As shown in Fig. 1, the QR factorization on the global weight is $\mathbf{X} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ an orthogonal matrix and $\mathbf{R}$ is an upper triangular matrix. The projection of the local gradient in the direction spanned by the new basis matrix $\mathbf{Q}$ ensures that weight updates from individual clients align with the overall direction of the global model while preserving acquired knowledge. In special case when the global model $\mathbf{x}_{t-1}$ and the current gradient $\nabla F_{i,t}(\mathbf{x}, \xi)$ are similar, our method benefit from stability propriety of $\mathbf{R}$. For simplicity, we will omit the index of the iteration round t . Given the factorization $\mathbf{X} = \mathbf{Q}\mathbf{R}$, then we have:

$$\mathbf{Q}^T \mathbf{X} = \mathbf{Q}^T \mathbf{Q} \mathbf{R}$$

and since, $\mathbf{I} = \mathbf{Q}^\top \mathbf{Q}$, where $\mathbf{I}$ is an identity matrix it follows that:

$$\mathbf{Q}^\top \mathbf{X} = \mathbf{I} \mathbf{R} = \mathbf{R}$$

if $\mathbf{X} \approx \nabla F_i(\mathbf{x}, \xi)$, then we have:

$$\mathbf{Q}^\top \mathbf{X} \approx \mathbf{Q}^\top \nabla F_i(\mathbf{x}, \xi) \approx \mathbf{R} \quad (9)$$

Our projection in Eq. 7 can be rewritten as:

$$\text{proj}(\mathcal{G}) = \underbrace{\mathbf{Q}^\top \nabla F_i(\mathbf{x}, \xi)}_{\mathbf{R}} \quad (10)$$

The introduction of an upper triangular matrix in the projection computation can effectively improve the training stability in our model.

Algorithm 1. **FEDQR** in FedAvg

Input: initial model $\mathbf{x}^0$, N total number of client, C client's participation ratio, T number of total rounds, $\eta > 0$ learning rates.

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1: for  $t = 1, 2, \dots, T$  do
2:   Sample device set  $\mathcal{S}_t \subseteq [N]$  with  $|\mathcal{S}_t| = CN$ 
3:   for  $i \in \mathcal{S}_t$  in parallel do
4:      $\mathbf{x}_i^{t,1} \leftarrow \mathbf{x}^{t-1}$ 
5:     Compute QR factorization:  $\mathbf{Q}_i^{t,1} \leftarrow \text{QRfactorization}(\mathbf{x}_i^{t,1})$ 
6:     for  $j = 1, 2, \dots, J_i$  do
7:       Compute local gradient:  $g_{i,j}^t \leftarrow \nabla F_{i,t}(\mathbf{x}, \xi)$ 
8:       Compute gradient projection:  $\text{proj}_{i,j}^t \leftarrow \mathbf{Q}_i^{t,1\top} g_{i,j}^t$ 
9:       Compute projected gradient:  $\tilde{g}_{i,j}^t \leftarrow g_{i,j}^t - \text{proj}_{i,j}^t$ 
10:      Update model:  $\mathbf{x}_i^{t,j} \leftarrow \mathbf{x}_i^{t,j} - \eta \tilde{g}_{i,j}^t$ 
11:    end for
12:  end for
13:   $\mathbf{x}^t \leftarrow \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \mathbf{x}_i^t$ 
14: end for
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FedQR for Stability. FedQR enhances stability by addressing the divergence of client models from global optimum. Initially, the direction of the global model parameters is altered to align with a new basis represented by the matrix $\mathbf{Q}$ (see Fig. 2). This adjustment ensures that local updates remain in proximity to local optimal solutions while retaining knowledge acquired from the global model. Particularly in non-i.i.d data scenarios, preserving crucial features learned from the global model becomes paramount. Each client heavily relies on its unique data distribution, potentially leading to model instability if global knowledge isn't properly incorporated. The orthonormality of the $\mathbf{Q}$ matrix ensures stability, as gradients are projected onto a basis that spans the subspace without deviating too far from the optimal solution. Moreover, FedQR mitigates optimization oscillation, a phenomenon exacerbated by non-i.i.d data, which can

impede training progress. By employing FedQR, faster and more stable convergence is achievable, especially in heterogeneous data environments. Furthermore, the unit vectors in the $\mathbf{Q}$ matrix act as normalizers, preventing the occurrence of vanishing or exploding gradients. As shown in Eq. 10, when the similarities between our local and global models are high, we introduce an upper triangular matrix $\mathbf{R}$ in the projection. This plays a crucial role in reducing our model’s sensitivity to noise by directing its focus towards the most pertinent features. By prioritizing the essential features from both the local and global models, we can avoid overfitting—an issue commonly associated with overparameterized models that tend to memorize training data, thereby limiting their performance on unseen data. In contrast to FedProx, which operates with full model parameters, our method also leverages the use of the upper triangular matrix, leading to more stable optimization. The FedQR algorithm, depicted in Algorithm 1, decomposes the global weight parameter into matrices $\mathbf{Q}$ and $\mathbf{R}$ (line 5). Utilizing the $\mathbf{Q}$ matrix, we proceed to compute the projected matrix and subsequently project the current local gradient onto the $\mathbf{Q}$ basis (lines 8–9). Finally, we update the model weights. Once all client weights are collected, the server performs an aggregation (line 13).

4 Results and Discussions

Our experiments involves both simulated i.i.d, label skew non-i.i.d data, and real-world feature shift data. The main results demonstrate that our method can accelerate convergence, and improve accuracy with no learning rate tuning.

4.1 Experiment Setup

We conducted our experiment on a MacOS system equipped with an M1 chip with 16 GB RAM. We first simulate three data distributions using the MNIST, FMNIST, and CIFAR-10 datasets: an i.i.d distribution with balanced data, and two non-i.i.d distributions. To induce non-i.i.dness, we utilized a Dirichlet distribution with concentration parameters set to $\alpha = 0.1$ and $\alpha = 0.05$. This distribution introduced label skewness in our datasets, where smaller values of α correspond to higher data heterogeneity. Then, we evaluate our method on real world feature shift non-i.i.d using Digit-5 dataset and Amazon-Review dataset. Each dataset was partitioned such that 75% of the total data was allocated for training, with the remainder reserved for testing. The training data was further divided among 10 clients to simulate a federated learning environment. For the image classification tasks involving MNIST, FMNIST, CIFAR-10, and Digit-5 datasets, we employed a convolutional neural network (CNN) architecture in [20]. For the text classification task using the Amazon-Review dataset, we opted for a multilayer perceptron (MLP) model [25]. We configured the local round and global round parameters to 1 and 100, respectively, as the default settings. However, in specific experiments, such as the Digit-5 scenario, we adjusted the global round parameter to 70. By default, the learning rate and batch size are

set to 0.005 and 10, respectively. In cases where the default learning rate did not yield satisfactory performance, we tuned the learning rate for each algorithm, except for our proposed method. This underscored the robustness of our method in learning rate sensitivity, as demonstrated by its consistent performance across different settings.

4.2 Datasets

In our empirical evaluation we use the following datasets:

- **MNIST**: The MNIST [13] dataset, known for handwritten digit classification, comprises 60,000 training images and 10,000 testing images. Each grayscale image features a resolution of 28×28 pixels, covering digits from 0 to 9, with a distribution of 6,000 training images and 1,000 testing images per digit.
- **CIFAR-10**: CIFAR-10 [12] consists of 50,000 training images and 10,000 test images divided into 10 classes. Each image has a size of 32×32 pixels and is colored in RGB. The dataset contains 5,000 training images and 1,000 testing images for each class.
- **Fashion-MNIST (FMNIST)**: FMNIST [24] offers a more advanced version of the MNIST dataset, moving from handwritten digits to include 10 different fashion items such as T-shirts, trousers, and bags. the training and test data are split similarly to MNIST. Each example comprises a grayscale image measuring 28×28 pixels, categorized into 10 different labels.
- **Digit-5**: Digit-5 gathers five common digit datasets, including MNIST [13], MNIST-M [5], Synthetic Digits [5], SVHN [21], and USPS [8]. In our setting, each dataset represents a client. These datasets together offer a comprehensive representation of digit images from different sources and domains, representing a rigorous evaluation of our methods on feature shifts non-i.i.d problem.
- **Amazon Review**: The Amazon review dataset [28] comprises customer reviews spanning various product categories, including Books, DVD, Electronics, Kitchen, and Houseware. Within this dataset, users provide ratings indicating their satisfaction levels with the products, ranging from positive to negative sentiments. In our experiment, each clients is responsible for one product categories.

4.3 Baseline Methods

To benchmark the performance of our approach, we compare it against the following baseline methods

- **FedAvg** [20]: FedAvg is the pioneering federated learning method, assuming i.i.d data and utilizing average aggregation.
- **FedGen** [31]: FedGen uses knowledge distillation to control client drift via synthetic data generated from clients.
- **SCAFFOLD** [11]: SCAFFOLD enhances training convergence by introducing two additional variables to mitigate client drift.

- **FedDyn** [1]: FedDyn aimed at minimizing communication overhead in non-i.i.d data scenarios, especially in cases of partial device participation.
- **FedBN** [18]: FedBN introduces a local batch normalization method to the original FedAvg and is designed to alleviate feature shift non-i.i.d data.

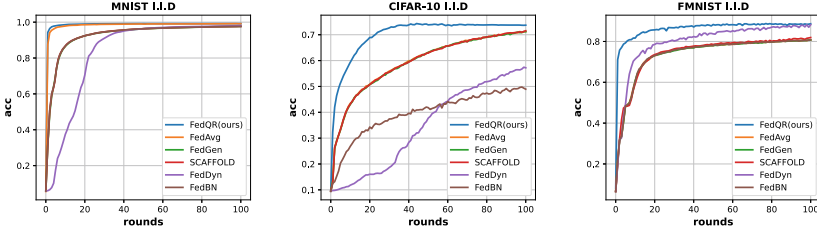


Fig. 3. Communication efficiency evaluation under i.i.d data

4.4 Communication Efficiency

Analysis for i.i.d Data: In Fig. 3, we observe that FedQR exhibits similar performance to FedAvg, with slightly better convergence in the initial stages on the MNIST dataset. In contrast, other methods show inferior convergence rates compared to FedAvg. Specifically, FedDyn demonstrates slow convergence during the earlier rounds. Under the CIFAR-10 homogeneous data scenario, even with learning rate tuning from the set $\{0.1, 0.01, 0.001, 0.0001\}$, FedQR still achieves the fastest convergence among all the algorithms, requiring only 40 rounds to converge. Similar results are observed for the FMNIST dataset, where the convergence rates of FedAvg, FedBN, FedGEN, and SCAFFOLD are lower than those of FedDyn and FedQR. Notably, our method achieves convergence in just 40 rounds, whereas FedDyn requires 100 rounds to converge.

Analysis for Non-i.i.d Data with $\alpha = 0.1$: As depicted in Fig. 4, our method demonstrates robustness in heterogeneous situations, significantly improving convergence and communication efficiency while achieving higher test accuracy compared to other methods when trained on MNIST datasets. Specifically, FedQR achieves 97% accuracy in just 36 rounds of training, whereas SCAFFOLD requires 100 rounds to reach a comparable result. In the case of CIFAR-10, FedAvg, FedDyn, SCAFFOLD, FedBN, and FedGEN all experienced significant drops in performance, even with learning rate tuning using values $\{0.1, 0.01, 0.001, 0.005, 0.0001\}$. The results clearly show that our method significantly reduces communication costs while achieving better test accuracy. FedQR achieves an accuracy of 41.5% in just 13 communication rounds, while FedDyn requires 100 rounds to attain comparable results, resulting in a 7.6% reduction in communication. Similarly, for FMNIST datasets, the results show reduced

performance, particularly for FedDyn and SCAFFOLD, possibly due to the complexity of finding optimal hyperparameters to control client drift effectively. In contrast, FedQR accelerates convergence by up to 5% compared to the other methods.

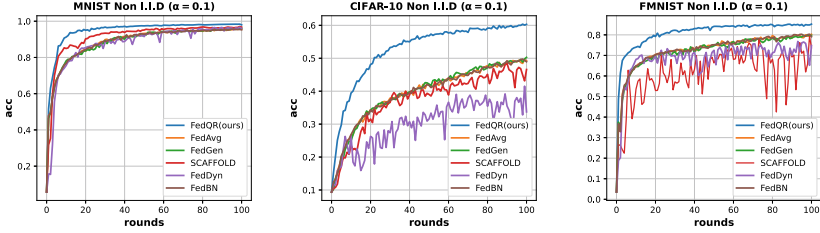


Fig. 4. Communication efficiency evaluation under non-i.i.d data ($\alpha = 0.1$)

Analysis for Non-i.i.d Data with $\alpha = 0.05$: Under non-i.i.d conditions ($\alpha = 0.005$), we observe fluctuating accuracy curves with lower performance for FedDyn and SCAFFOLD across all datasets, except in MNIST where FedDyn demonstrates faster convergence compared to vanilla FedAvg. On the other hand, our method accelerates communication fivefold relative to FedDyn and SCAFFOLD while achieving better test accuracy in CIFAR-10 and FMNIST datasets (Fig. 5).

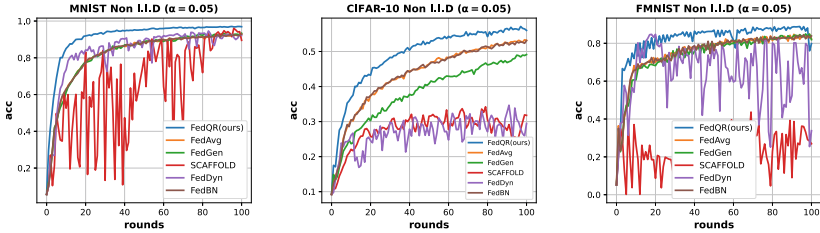


Fig. 5. Communication efficiency evaluation of MNIST, CIFAR-10, and FMNIST datasets under non-i.i.d data ($\alpha = 0.05$)

Analysis for Feature Shift Non-i.i.d Data: In Fig. 6, which examines feature shift data, the results show that FedQR can converge in just 20 rounds of training for the Digit-5 dataset and 10 rounds of training for the Amazon-review dataset. However, other algorithms display poor performance in terms of convergence and accuracy. Notably, FedBN, despite being designed to tackle feature shift, performs worse than FedQR. This illustrates the effectiveness of our method across various data distributions.

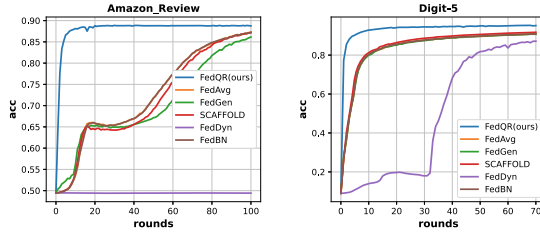


Fig. 6. Test accuracy convergence of Amazon Review and Digit-5 datasets

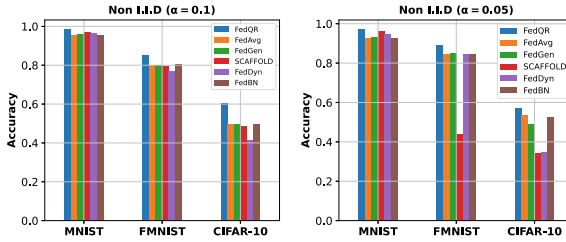


Fig. 7. Performance comparison of test accuracy under non-i.i.d with $\alpha = 0.1$ to the left, and $\alpha = 0.05$ to the right

4.5 Test Accuracy Analysis

Figure 7 illustrates the test accuracy of the MNIST, FMNIST, and CIFAR-10 datasets under both non-i.i.d settings with $\alpha = 0.1$ and $\alpha = 0.05$. Under the non-i.i.d setting with $\alpha = 0.1$, the results demonstrate that FedQR consistently outperforms other algorithms across all datasets. Interestingly, as the dataset complexity increases, our method show even higher performance, particularly with the FMNIST and CIFAR-10 datasets. When we further increase data heterogeneity to $\alpha = 0.05$ (Fig. 7 right), we observe that FedQR maintains its superior accuracy compared to other algorithms. For the simpler MNIST dataset, SCAFFOLD achieves the highest accuracy among the methods. However, as we transition to more complex datasets such as FMNIST and CIFAR-10, SCAF-

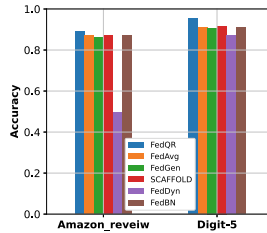


Fig. 8. Performance comparison of test accuracy of Amazon Review and Digit-5 datasets

FOLD’s accuracy performance declines significantly. In contrast, FedAvg and FedGen demonstrate better adaptability to overcome the complexities of heterogeneous data, maintaining competitive performance across all datasets. In Fig. 8, we present the test accuracy of the methods applied to both the Amazon-Review and Digit-5 datasets. Our results indicate that our proposed method have competitive performance compared to FedAvg, FedBN, and SCAFFOLD, while significantly outperforming FedDyn across both datasets. Specifically, for the Amazon-Review dataset, our method achieves comparable results to FedAvg, FedBN, and SCAFFOLD. However, it surpasses FedDyn by a considerable margin. In the case of the Digit-5 dataset, our method demonstrates superior performance, followed closely by SCAFFOLD. FedAvg, FedGEN, and FedBN has similar levels of performance, while FedDyn shows the lowest accuracy. These results demonstrates the efficacy of our method in improving accuracy, particularly in the context of feature shift non-i.i.d data.

4.6 Computation Costs

To evaluate the computational cost of FedQR, we report the execution time per round and the total execution time in Table 1. The results indicate that our method completes faster by 2.3s, 6.23s, 6.45s, and 22.7s compared to SCAFFOLD, FedDyn, FedBN, and FedGEN, respectively, demonstrating a lower computational cost. Furthermore, among all the compared methods, only FedQR and FedBN do not require additional hyperparameters.

Table 1. Execution time (↓) on FMNIST and the use of additional hyperparameters

Algorithm	Total Time	Time/Iteration	Additional hyperparameter
FedDyn	2699.94 s	26.76 s	✓
SCAFFOLD	2302.94 s	22.83 s	✓
FedGen	4367.31 s	43.23 s	✓
FedBN	2725.99 s	26.98 s	✗
FedQR (Ours)	2073.99 s	20.53 s	✗

4.7 Analysis in Partial Client Participation

Our experiments on partial client participation, where we have 40% of the total clients drop in each round of communication is illustrated in Fig. 9. For non-i.i.d with $\alpha = 0.1$, the findings reveal a notable underperformance of the SCAFFOLD and FedDyn algorithms. These methods show unstable convergence curves accompanied by low accuracy levels. Despite FedGEN demonstrating slow convergence initially, its performance aligns closely with that of FedAvg and

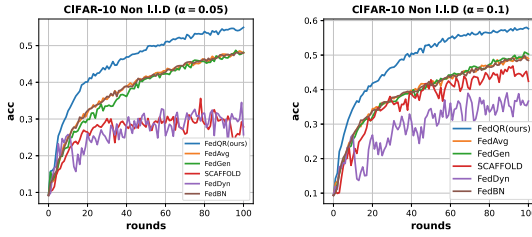


Fig. 9. Communication efficiency evaluation for CIFAR-10 with non-i.i.d data distribution in partial client participation: Left illustrates the convergence under $\alpha = 0.1$, while the right represents convergence under $\alpha = 0.05$

FedBN in the later stages of training. However, it is evident that FedQR consistently outperforms all other algorithms. This illustrates the robustness of FedQR in partial client participations.

In the case of non-i.i.d data with $\alpha = 0.05$ (Fig. 9 right), we observe that FedDyn exhibits the slowest convergence rate among the algorithms studied. SCAFFOLD, on the other hand, demonstrates a slower convergence curve compared to FedAvg, FedGEN, and FedBN. In contrast, our method, FedQR, demonstrates a clear ability to accelerate learning. In fact, it achieves a convergence rate twice as fast as the best-performing algorithm among the other comparative algorithms. These results clearly indicate the efficiency of our method in reducing communication overhead, particularly in scenarios involving partial client participation.

5 Conclusion

We propose a new reparameterization method via QR factorization to overcome unstable and slow convergence in heterogeneous federated learning. Our method is simple yet can converge in fewer communication rounds without being sensitive to learning rate tuning. Extensive experiments demonstrate that FedQR is well suited to both full and partial client participation, especially in non-i.i.d data. Future works will focus on extending FedQR to more stable factorization methods and performing both practical and theoretical convergence analyses in large-scale heterogenous federated learning settings.

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References

1. Acar, D.A.E., Zhao, Y., Navarro, R.M., Mattina, M., Whatmough, P.N., Saligrama, V.: Federated learning based on dynamic regularization. arXiv preprint [arXiv:2111.04263](https://arxiv.org/abs/2111.04263) (2021)
2. Chang, H., Shejwalkar, V., Shokri, R., Houmansadr, A.: Cronus: robust and heterogeneous collaborative learning with black-box knowledge transfer. arXiv preprint [arXiv:1912.11279](https://arxiv.org/abs/1912.11279) (2019)
3. Cheney, W., Kincaid, D.: Linear algebra: theory and applications. Aust. Math. Soc. **110**, 544–550 (2009)
4. Farajtabar, M., Azizan, N., Mott, A., Li, A.: Orthogonal gradient descent for continual learning (2019). <https://arxiv.org/abs/1910.07104>
5. Ganin, Y., Lempitsky, V.: Unsupervised domain adaptation by backpropagation. In: Proceedings of the 32nd International Conference on International Conference on Machine Learning, ICML 2015, vol. 37, pp. 1180–1189. JMLR.org (2015)
6. Garg, D., Alam, M.: Deep learning and IoT for agricultural applications. In: Alam, M., Shakil, K.A., Khan, S. (eds.) Internet of Things (IoT), pp. 273–284. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-37468-6_14
7. Householder, A.S.: Unitary triangularization of a nonsymmetric matrix. J. ACM **5**(4), 339–342 (1958). <https://doi.org/10.1145/320941.320947>
8. Hull, J.: A database for handwritten text recognition research. IEEE Trans. Pattern Anal. Mach. Intell. **16**(5), 550–554 (1994). <https://doi.org/10.1109/34.291440>
9. Hyeon-Woo, N., Ye-Bin, M., Oh, T.H.: Fedpara: low-rank hadamard product for communication-efficient federated learning (2023)
10. Jeong, W., Yoon, J., Yang, E., Hwang, S.J.: Federated semi-supervised learning with inter-client consistency & disjoint learning (2021). <https://arxiv.org/abs/2006.12097>
11. Karimireddy, S.P., Kale, S., Mohri, M., Reddi, S., Stich, S., Suresh, A.T.: Scaffold: stochastic controlled averaging for federated learning. In: International Conference on Machine Learning, pp. 5132–5143. PMLR (2020)
12. Krizhevsky, A., Hinton, G., et al.: Learning multiple layers of features from tiny images. Master’s thesis, Department of Computer Science, University of Toronto (2009)
13. Lecun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. Proc. IEEE **86**(11), 2278–2324 (1998). <https://doi.org/10.1109/5.726791>
14. Li, C., Farkhoor, H., Liu, R., Yosinski, J.: Measuring the intrinsic dimension of objective landscapes (2018). <https://arxiv.org/abs/1804.08838>
15. Li, D., Wang, J.: Fedmd: heterogenous federated learning via model distillation. arXiv preprint [arXiv:1910.03581](https://arxiv.org/abs/1910.03581) (2019)
16. Li, T., Sahu, A.K., Zaheer, M., Sanjabi, M., Talwalkar, A., Smith, V.: Federated optimization in heterogeneous networks. In: Dhillon, I., Papailiopoulos, D., Sze, V. (eds.) Proceedings of Machine Learning and Systems, vol. 2, pp. 429–450 (2020). https://proceedings.mlsys.org/paper_files/paper/2020/file/1f5fe83998a09396eb6477d9475ba0c-Paper.pdf
17. Li, W., et al.: A comprehensive survey on machine learning-based big data analytics for IoT-enabled smart healthcare system. Mob. Netw. Appl. **26**(1), 234–252 (2021). <https://doi.org/10.1007/s11036-020-01700-6>
18. Li, X., Jiang, M., Zhang, X., Kamp, M., Dou, Q.: Fedbn: federated learning on non-IID features via local batch normalization (2021)

19. Ma, X., Yu, H., Wang, Y., Wang, Y.: Large-scale transportation network congestion evolution prediction using deep learning theory. *PLOS ONE* **10**(3), e0119044 (2015). <https://doi.org/10.1371/journal.pone.0119044>
20. McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: *Artificial Intelligence and Statistics*, pp. 1273–1282. PMLR (2017)
21. Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A.Y.: Reading digits in natural images with unsupervised feature learning. In: *NIPS Workshop on Deep Learning and Unsupervised Feature Learning 2011* (2011). http://ufldl.stanford.edu/housenumbers/nips2011_housenumbers.pdf
22. Reddi, S., et al.: Adaptive federated optimization (2021)
23. Roberts, D.A.O., Roberts, L.R.: QR and LQ decomposition matrix backpropagation algorithms for square, wide, and deep – real or complex – matrices and their software implementation (2020)
24. Xiao, H., Rasul, K., Vollgraf, R.: Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms (2017)
25. Yang, L., Balaji, Y., Lim, S.-N., Shrivastava, A.: Curriculum manager for source selection in multi-source domain adaptation. In: Vedaldi, A., Bischof, H., Brox, T., Frahm, J.-M. (eds.) *ECCV 2020*. LNCS, vol. 12359, pp. 608–624. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58568-6_36
26. Yao, D., et al.: Fedhm: efficient federated learning for heterogeneous models via low-rank factorization. arXiv preprint [arXiv:2111.14655](https://arxiv.org/abs/2111.14655) (2021)
27. Yoon, J., Jeong, W., Lee, G., Yang, E., Hwang, S.J.: Federated continual learning with weighted inter-client transfer. In: *International Conference on Machine Learning*, pp. 12073–12086. PMLR (2021)
28. Zhang, X., Zhao, J., LeCun, Y.: Character-level convolutional networks for text classification. In: Cortes, C., Lawrence, N., Lee, D., Sugiyama, M., Garnett, R. (eds.) *Advances in Neural Information Processing Systems*, vol. 28. Curran Associates, Inc. (2015). https://proceedings.neurips.cc/paper_files/paper/2015/file/250cf8b51c773f3f8dc8b4be867a9a02-Paper.pdf
29. Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D., Chandra, V.: Federated learning with non-IID data. arXiv preprint [arXiv:1806.00582](https://arxiv.org/abs/1806.00582) (2018)
30. Zhou, H., Cheng, J., Wang, X., Jin, B.: Low rank communication for federated learning. In: Nah, Y., Kim, C., Kim, S.H., Moon, Y.-S., Whang, S.E. (eds.) *DAS-FAA 2020*. LNCS, vol. 12115, pp. 1–16. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-59413-8_1
31. Zhu, Z., Hong, J., Zhou, J.: Data-free knowledge distillation for heterogeneous federated learning. In: *International Conference on Machine Learning*, pp. 12878–12889. PMLR (2021)